UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 09/06/2022*

PROBLEM

We observe N samples of a realization of the random process $X[n] = A + W[n] + \frac{1}{2}W[n-1]$, for n=0, 1,, N-1. where A is modelled as a Gaussian random variable with zero mean and variance σ_A^2, and $W[n]$ is a white Gaussian random process with variance σ_W^2. Let us define the signal to noise ratio (SNR) as follows:

$$\gamma \triangleq \frac{E\left\{A^2\right\}}{E\left\{\left(W[n] + \frac{1}{2}W[n-1]\right)^2\right\}}.$$

(1) Derive the autocorrelation function (ACF) $R_X[m] = E\left\{X[n]X[n+m]\right\}$ and the power spectral density (PSD) $S_X\left(e^{j2\pi f}\right) = FT\left\{R_X[m]\right\}$ of the observed signal $X[n]$ and plot them.

(2) Derive the optimal estimator, i.e. MMSE estimator, of A given the observation of $X[n]$ and $X[n-1]$ (i.e. we observe N=2 samples) and express it as a function of γ. Show that the estimator can be expressed as $\hat{A}_{MMSE} = \alpha(\gamma)\bar{X}$, where α is a scalar function of the SNR γ and $\bar{X}$ is the sample mean of the observed data.

(3) Derive the mean square error (MSE) of $\hat{A}_{MMSE}$ as a function of σ_A^2 and γ.

(4) Assuming σ_A^2=1, plot the MSE as a function of γ and find the minimum value of γ which guarantees an MSE lower than 10^{-3}; finally analyze and comment the behavior of the estimator $\hat{A}_{MMSE}$ and of the MSE when γ tends to 0 or $+\infty$.

Solution

(1) The ACF of $X[n]$ is instead given by:

$$\begin{aligned} R_X[m] &= E\{X[n]X[n+m]\} = E\left\{\left(A+W[n]+\frac{1}{2}W[n-1]\right)\left(A+W[n+m]+\frac{1}{2}W[n+m-1]\right)\right\} \\ &= E\{A^2\}+E\left\{\left(W[n]+\frac{1}{2}W[n-1]\right)\left(W[n+m]+\frac{1}{2}W[n+m-1]\right)\right\} \\ &= \sigma_A^2+R_X[m]+\frac{1}{4}R_X[m]+\frac{1}{2}R_X[m-1]+\frac{1}{2}R_X[m+1] \\ &= \sigma_A^2+\frac{5}{4}\sigma_W^2\delta[m]+\frac{1}{2}\sigma_W^2\delta[m-1]+\frac{1}{2}\sigma_W^2\delta[m+1] \\ &= \sigma_W^2\left(\gamma+\frac{5}{4}\delta[m]+\frac{1}{2}\delta[m-1]+\frac{1}{2}\delta[m+1]\right) \end{aligned}$$

The PSD is the Fourier Transform (FT) of the ACF:

$$\begin{aligned} S_X\left(e^{j2\pi f}\right) &= FT\{R_X[m]\} \\ &= 2\pi\sigma_A^2\delta\left(e^{j2\pi f}-1\right)+\frac{5}{4}\sigma_W^2+\frac{1}{2}\sigma_W^2 e^{-j2\pi f}+\frac{1}{2}\sigma_W^2 e^{j2\pi f} \\ &= 2\pi\sigma_A^2\delta\left(e^{j2\pi f}-1\right)+\frac{5}{4}\sigma_W^2+\sigma_W^2\cos(2\pi f) \\ &= \sigma_W^2\left(2\pi\gamma\delta\left(e^{j2\pi f}-1\right)+\frac{5}{4}+\cos(2\pi f)\right) \end{aligned}$$

(2) Since the observed data and the parameter to be estimated are jointly Gaussian, the MMSE estimator of A given the observed data $X[n]$ and $X[n-1]$ coincides with the MAP estimator and with the LMMSE estimator. The LMMSE estimator can be derived by imposing that the estimator is linear, unbiased and satisfies the Orthogonality principle:

$$\hat{A}_{LMMSE} = \mathbf{a}^T\mathbf{X}+b = a_1X[n]+a_2X[n-1]+b, \text{ where } \mathbf{X}=\begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix}$$

$$\varepsilon = A-\hat{A}_{LMMSE} = A-a_1X[n]-a_2X[n-1]-b$$

$$E\{\varepsilon\} = E\{A-a_1X[n]-a_2X[n-1]-b\} = -b = 0 \quad \Rightarrow \quad b=0$$

$$E\{\varepsilon X[n-m]\} = E\{(A - a_1 X[n] - a_2 X[n-1])\, X[n-m]\} = 0, \quad m = 0,1$$

$$= E\{AX[n-m]\} - a_1 E\{X[n]X[n-m]\} - a_2 E\{X[n-1]X[n-m]\} = 0, \quad m = 0,1$$

$$= E\{A^2\} - a_1 R_X[m] - a_2 R_X[m-1] = 0, \quad m = 0,1$$

$$= \sigma_A^2 - a_1 R_X[m] - a_2 R_X[m-1] = 0, \quad m = 0,1$$

$$\begin{cases} a_1 R_X[0] + a_2 R_X[-1] = \sigma_A^2 \\ a_1 R_X[1] + a_2 R_X[0] = \sigma_A^2 \end{cases}$$

$$\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix} \mathbf{a} = \sigma_A^2 \mathbf{1}$$

$$\mathbf{a} = \begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix}^{-1} \sigma_A^2 \mathbf{1} = \frac{1}{\sigma_W^2} \begin{bmatrix} \gamma + \frac{5}{4} & \gamma + \frac{1}{2} \\ \gamma + \frac{1}{2} & \gamma + \frac{5}{4} \end{bmatrix}^{-1} \sigma_A^2 \mathbf{1} = \gamma \begin{bmatrix} \gamma + \frac{5}{4} & \gamma + \frac{1}{2} \\ \gamma + \frac{1}{2} & \gamma + \frac{5}{4} \end{bmatrix}^{-1} \mathbf{1}$$

$$= \frac{\gamma}{\left(\gamma + \frac{5}{4}\right)^2 - \left(\gamma + \frac{1}{2}\right)^2} \begin{bmatrix} \gamma + \frac{5}{4} & -\gamma - \frac{1}{2} \\ -\gamma - \frac{1}{2} & \gamma + \frac{5}{4} \end{bmatrix} \mathbf{1} = \frac{\frac{3}{4}\gamma}{\left(2\gamma + \frac{7}{4}\right)\frac{3}{4}} \mathbf{1} = \frac{\gamma}{\left(2\gamma + \frac{7}{4}\right)} \mathbf{1} = \frac{4\gamma}{8\gamma + 7} \mathbf{1}$$

$$a_1 = a_2 = \frac{4\gamma}{8\gamma + 7}$$

The fact that $a_1 = a_2$ could have been said from the very beginning since the correlation between the parameter to be estimated A and the two data samples are the same, i.e. $E\{AX[n-m]\} = \sigma_A^2, \quad m = 0,1$.

$$\hat{A}_{LMMSE} = \frac{4\gamma}{8\gamma+7}X[n] + \frac{4\gamma}{8\gamma+7}X[n-1] = \frac{8\gamma}{8\gamma+7}\left(\frac{X[n]+X[n-1]}{2}\right) = \alpha\bar{X}$$

where $\alpha \triangleq \frac{8\gamma}{8\gamma+7}$ and $\bar{X} \triangleq \frac{X[n]+X[n-1]}{2}$

(3) MSE of the LMMSE estimator:

$$\begin{aligned} MSE &= E\left\{\varepsilon^2\right\} = E\left\{\varepsilon A\right\} = E\left\{\left(A - a_1 X[n] - a_2 X[n-1]\right)A\right\} \\ &= \sigma_A^2 - a_1 E\left\{X[n]A\right\} - a_2 E\left\{X[n-1]A\right\} \\ &= \sigma_A^2\left(1 - a_1 - a_2\right) = \sigma_A^2\left(1 - \frac{\alpha}{2} - \frac{\alpha}{2}\right) = \sigma_A^2\left(1-\alpha\right) \\ &= \sigma_A^2\left(1 - \frac{8\gamma}{8\gamma+7}\right) = \frac{7\sigma_A^2}{8\gamma+7} \end{aligned}$$

(4) MSE as a function of γ when σ_A^2=1:

$$MSE = \left.\frac{7\sigma_A^2}{8\gamma+7}\right|_{\sigma_A^2=1} = \frac{7}{8\gamma+7} \le 10^{-3} \quad \Rightarrow \quad \gamma \ge \frac{7\cdot 10^3 - 7}{8} = 874.125$$

$$\hat{A}_{LMMSE} = \alpha(\gamma)\bar{X}, \quad MSE = \sigma_A^2\left(1-\alpha(\gamma)\right) = \frac{7\sigma_A^2}{8\gamma+7}, \quad \alpha(\gamma) = \frac{8\gamma}{8\gamma+7}$$

$$\gamma \to +\infty \quad \Rightarrow \quad \alpha(\gamma) \to 1 \quad \Rightarrow \quad \hat{A}_{LMMSE} \to \bar{X}, MSE \to 0$$

$$\gamma \to 0 \quad \Rightarrow \quad \alpha(\gamma) \to 0 \quad \Rightarrow \quad \hat{A}_{LMMSE} \to 0 = E\left\{A\right\}, MSE \to \sigma_A^2$$